Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 70%. We keep your highest score.

Next item →

1. In how many ways can 3 students be selected from a group of 5 to stand in line? 1 / 1 point

- ○ 10
- ○ 20
- ◉ 60
- ○ 30

Nice work

The answer is $P(5,3) = 60$. You can also solve it by calculating $5 \times 4 \times 3 = 20$

2. Assume there are 4 different pop (P) songs and 4 different classical(C) songs, and you want to listen to them in an alternate way. In how many ways can you do that? 1 / 1 point

- ○ 144
- ○ 256
- ○ 576
- ◉ 1152

Nice work

Note that its either $CPCPCPCP$ or $PCPCPCPC$. For each case there are $4! \times 4! = 576$ different ways to listen. So the answer is 2×576

3. Consider the following set $\{1, 2, \cdots, 10\}$. In how many ways can we pick 4 numbers such that two of them are even and two of them are odd. 1 / 1 point

- ○ 80
- ○ 112
- ○ 120
- ◉ 100

Nice work

There are 5 odd numbers and 5 even numbers. The answer is $\binom{5}{2} \times \binom{5}{2} = 10 \times 10 = 100$

4. Consider the following two sets $A = \{1, 2, 3, 4\}$ $B = \{a, b, c, d, e\}$. In how many ways we can create a 4-character password such that it contains two elements from A and two elements from B? 1 / 1 point

- ◉ 1440
- ○ 480
- ○ 960
- ○ 240

Nice work

We have $\binom{5}{2}$ ways to pick two elements from B and $\binom{4}{2}$ ways to pick two elements from A. They also can form 4! permutations. So the answer is $\binom{5}{2} \times \binom{4}{2} \times 4!$.

5. 5 countries participate in a competition by each sending 10 athletes. In how many ways we can pick 3 athletes such that they all belong to different countries. 1 / 1 point

- ○ 10^6
- ○ 10^5
- ○ 10^3
- ◉ 10^4

Nice work

Note that we first must pick 3 countries from 5 which can be done in $\binom{5}{3}$ different ways. Now from each country, there are 10 ways to pick an athlete. So the answer is $\binom{5}{3} \times 10^3 = 10^4$.

6. How many ways can we arrange the letters in the word ``SUCCESS"? 1 / 1 point

- ○ $\frac{7!}{2! \times 2!}$
- ○ $7!$
- ◉ $\frac{7!}{2! \times 3!}$
- ○ $\frac{7!}{2}$

Nice work

There are $7!$ permutations in total. However, as there are two ``C" and three ``S", each case appears $2! \times 3!$ times. So we must divide $7!$ by $2! \times 3!$.

7. Consider the previous question, but now there are only 3 different pop songs and 4 different classical songs. Assume you want to listen to them in an alternate way. In how many ways can you do that? 1 / 1 point

- ○ $3! \times 3!$
- ○ $2 \times 3! \times 3!$
- ○ $2 \times 3! \times 4!$
- ◉ $3! \times 4!$

Nice work

Note that now the order can only be $CPCPCPC$. So the answer is $3! \times 4!$.

8. Assume the CEO, CTO and 4 employees are planning to sit around a round table. In how many ways the CEO and CTO are sitting exactly in front of each other. 1 / 1 point

- ◉ 24
- ○ 40
- ○ 60
- ○ 48

Nice work

We know that these 6 people have $5!$ ways to sit around a table. Consider the CEO to be seated anywhere. The CTO, has 5 places to sit, however in only one of them he is in front of the CEO. So the answer is $\frac{5!}{5} = 24$

9. Assume 5 people $\{a, b, c, d, e\}$ are sitting next to each other in a line. In how many ways a and b are sitting next to each other? 1 / 1 point

- ○ $5! \cdot 2!$
- ○ $5!$
- ○ $\frac{5!}{2!}$
- ◉ $4! \cdot 2!$

Nice work

We can assume a, b are attached to each other and form a ``pack". They have $2!$ permutations among themselves. And now there are 3 other people and their ``pack" and they have 4! Permutations. So the answer is $4! \times 2!$.

10. Assume $\binom{n}{8} = 2\binom{n}{7}$. What is the value of n? 1 / 1 point

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Nice work